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A Standardized Pipeline for Top-Down Spatial Biodiversity Modeling

Caleb Caulk

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Department of Information Technology
Uppsala University
Sweden

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Contents

1 Introduction

1.1 Motivation

Biodiversity is defined as "the variability among living organisms from all sources including, inter alia, terrestrial, marine and other aquatic ecosystems and the ecological complexes of which they are part; this includes diversity within species, between species and of ecosystems."[1] In this sense, biodiversity refers to the biotic component of ecological systems: the variety and variability of living organisms, from plants and animals to fungi, bacteria, and protists. Biodiversity is integral to ecosystem functioning and stability, supporting processes such as nutrient cycling, pollutant breakdown, water regulation, carbon storage, and ecosystem productivity (Stork 2009; Cardinale et al. 2012; Gascon et al. 2015). Furthermore, biodiversity is fundamental to human survival and well-being. It supports food security through pollination, soil fertility, fisheries, and livestock systems (Gascon et al. 2015; Brauman et al. 2020). It also provides biological materials and natural compounds used in medicine and drug discovery, including antibiotics such as penicillin (Gascon et al. 2015). Beyond these material contributions, biodiversity supports regulating ecosystem services such as water purification, climate regulation, and carbon storage, while also contributing to biomass-based energy systems and sustaining human health, livelihoods, and well-being (Brauman et al. 2020; Naeem et al. 2016; Kim et al. 2024).

Despite the overwhelming benefits of biodiversity to humanity and ecosystems, biodiversity is declining at an unprecedented rate. Approximately 74% of the world's terrestrial wilderness has experienced degradation, and approximately 9.6% has been completely lost since the early 1990s (Watson et al. 2016). Marine ecosystems have also been heavily affected, with 66% of the ocean's surface having been significantly impacted by human activities (Díaz et al. 2019). Moreover, it is reasonably likely that most coral reefs will be lost if global temperatures exceed 2°C (Frieler et al. 2012). Furthermore, human activities threaten the survival of approximately 1 million species, with approximately 25% of plants and animals threatened (Díaz et al. 2019). Species extinction rates have also increased and are now estimated to be around 1,000 times higher than the natural background rate (Pimm et al. 2014). Compounding this challenge, conservation efforts are underfunded, with a deficit exceeding half a trillion US dollars annually (Deutz et al. 2020). The acceleration of biodiversity loss and the drastic underfunding of conservation efforts have necessitated a shift towards the prioritization of biodiversity conservation and ecological rehabilitation.

Many initiatives around the world are being adopted to combat biodiversity loss and climate change, which are intricately linked. These initiatives focus on international cooperation, expanding protected areas, reducing pollution, and providing incentives for sustainable practices (IPBES 2019; IPCC 2023). Specific policy initiatives such as the 30x30 target—which aims to protect 30% of the world's land area by 2030—represent critical efforts for ecological recovery. If successful, they could generate large gains and rebounds in biodiversity. Achieving these goals would have significant impacts, including carbon sequestration, improved water quality regulation, and expanded capabilities to mitigate nutrient pollution (Zeng, Koh, and Wilcove 2022). For these initiatives to be successful, it is crucial to have accurate and robust models that can predict biodiversity patterns and trends. These models can inform conservation strategies, identify priority areas for protection, and assess the potential impacts of different conservation actions (Sinclair, White, and Newell 2010).

At their core, biodiversity models are spatial models that relate biodiversity patterns to environmental conditions at particular locations (Ferrier and Guisan 2006). In this thesis, we focus on community-level biodiversity models (Nieto-Lugilde et al. 2018), which refer to models that predict community-level biodiversity metrics, such as species richness, total abundance or relative abundance, intactness, or compositional similarity (Lamb et al. 2009; Santini et al. 2017). Broadly, community-level biodiveisty modeling approaches can be grouped into three main strategies based on how they produce predictions for community-level biodiversity metrics (Ferrier and Guisan 2006): (1) Bottom-up ("predict first, assemble later") approaches, which commonly use species distribution models (SDMs) to predict the spatial distribution of individual species based on environmental variables; multiple SDMs can be aggregated or stacked to estimate community-level metrics (Ferrier and Guisan 2006; Distler et al. 2015; Beery et al. 2021); (2) "assemble and predict together" approaches, which frequently use joint species distri-

[1] Convention on Biological Diversity (1993) *Article 2. Use of terms.* Available at: https://www.cbd.int/convention/articles/?a=cbd-02 (Accessed: 8 April 2026).

bution models (JSDMs) to model multiple species simultaneously, accounting for shared environmental responses and residual species associations (Warton et al. 2015; Wilkinson et al. 2019); and (3) top-down ("assemble first, predict later") approaches, in which the model directly predicts community-level biodiversity metrics rather than first modeling individual species or assemblages (Ferrier and Guisan 2006; Pollock et al. 2020).

While each strategy has its own strengths and weaknesses and excels in different contexts, it is unlikely that one modeling strategy will consistently outperform the others (Ferrier and Guisan 2006; Zhang et al. 2019). However, bottom-up approaches, are the most extensively researched modeling strategy for community-level biodiversity prediction. Many studies have focused on improving species distribution models, including benchmark evaluations (Norberg et al. 2019; Valavi et al. 2022), reviews of specific models (Waldock et al. 2022), and guidance on best practices (Araújo et al. 2019). Similarly, "assemble and predict together" models have also been the focus of many studies. Joint species distribution models have been developed partly to address the limited treatment of species co-occurrence and biotic interactions in traditional species distribution models (Wilkinson et al. 2019; Escamilla Molgora et al. 2022). Top-down models, on the other hand, are less widely studied and underutilized (Pollock et al. 2020). A significant gap remains in our understanding, comparison, and evaluation of bottom-up and top-down models, both between these modeling approaches and within the different methods they encompass. This highlights the need for further research and methodological development of top-down models (Ferrier and Guisan 2006; Pollock et al. 2020).

Currently, top-down models are often evaluated in individual studies that use different datasets, model types, biodiversity metrics, and evaluation procedures (Mondal and Bhat 2021; Cai et al. 2023; Ngor et al. 2023; Baggström et al. 2025). This can make it difficult to understand the current landscape of top-down models. While there have been some reviews of community-level models as a whole (Nieto-Lugilde et al. 2018), there still remains a lack of best practice guidelines for top-down models, comparable to those available for bottom-up approaches (Araújo et al. 2019). This has prompted the need for a standardized pipeline for models using a top-down strategy to allow for more rigorous testing (Pollock et al. 2020). This pipeline would allow for a more systematic evaluation of different top-down models and provide insights into their strengths and weaknesses in predicting biodiversity patterns and trends. Furthermore, a standardized pipeline would enable a more nuanced evaluation of these models and their applicability across different ecological contexts, and allow for quick comparison across different models. This would ultimately contribute to the advancement of biodiversity modeling as a whole and inform conservation strategies and policy decisions.

1.2 Research Objectives and Scope

The primary objective of this master's thesis is to develop a standardized pipeline for top-down modeling strategies that predict community-level biodiversity metrics and to provide a case study demonstrating the applicability and flexibility of this pipeline. To achieve this objective, a pipeline is first developed that integrates generalized data preprocessing, feature engineering, model training, and evaluation steps for top-down models. This pipeline is designed to be flexible and adaptable to a wide range of datasets and modeling approaches. Next, a case study is conducted using this pipeline to compare the performance of different models on derived biodiversity metrics from the United Kingdom Butterfly Monitoring Scheme (UKBMS) dataset (UK Butterfly Monitoring Scheme 2026), accessed through the Global Biodiversity Information Facility (GBIF)[2]. Models used in this case study will include statistical models, specifically generalized linear models and generalized additive models (Nelder and Wedderburn 1972; T. J. Hastie and Tibshirani 1986), as well as machine learning models, including random forests, gradient boosting, and artificial neural networks (Breiman 2001a; Friedman 2001; Lek et al. 1996). These results are then analyzed to provide insights into the strengths and weaknesses of different top-down models across different contexts. By applying the pipeline across contexts that vary in spatial scale, environmental predictor sets, and evaluation strategy, this thesis demonstrates its flexibility and general applicability to a range of community-level biodiversity modeling scenarios. This comparative framework also enables a systematic assessment of how different modeling classes (statistical or machine learning) perform under varying analytical conditions, thereby providing insight into their relative strengths, limitations, and appropriate use cases. The key research questions that this master's thesis aims to address are:

[2]Global Biodiversity Information Facility (GBIF). Available at: https://www.gbif.org/ (Accessed: 26 April 2026).

1. How do models perform with different strategies of cross-validation, specifically random cross-validation vs. spatial cross-validation?
2. How do models perform in data-limited scenarios?
3. How do different model classes (statistical vs. machine learning) perform compared to each other?

The results are therefore intended to illustrate the flexibility and practical utility of the pipeline, while demonstrating how standardized workflows can support more efficient and reproducible evaluation of top-down biodiversity models.

The remainder of this thesis is structured as follows. Section 2 provides background on biodiversity, including how it is defined and measured in Section 2.1, and reviews the main modeling approaches used to predict biodiversity patterns and trends in Section 2.2. Section 3 describes the methodological framework, including the pipeline design in Section 3.1, the case study dataset in Section 3.2, and the models implemented in Section 3.3. Section 4 presents the results of the case study and illustrates the flexibility of the proposed pipeline. Section 5 discusses the implications of these results for biodiversity modeling and conservation, while Section 5.2 addresses relevant ethical considerations. Section 6 outlines potential future directions and applications for research in this area. Finally, Section 7 concludes the thesis by summarizing the key findings and their implications for biodiversity modeling and conservation.

2 Background

This section introduces biodiversity and how it is measured in Section 2.1. Section 2.2 provides a mathematical framework for understanding top-down modeling approaches, so we can use them for predicting community-level biodiversity metrics. This background section provides the necessary context for understanding the methods and results presented in this thesis. It also highlights the differences between the modeling approaches as well as the different assumptions and limitations of these models.

2.1 Biodiversity and how to measure it

Biodiversity is a complex and multifaceted concept that encompasses the variety of life on Earth, including the diversity of species, genetic diversity within species, and the diversity of ecosystems (Stork 2009). Humanity has relied on biodiversity for survival for thousands of years as it provides us with food, materials, medicine, water security, and has a substantial impact on human health and well-being (Gascon et al. 2015; Naeem et al. 2016; Brauman et al. 2020; Kim et al. 2024). Large losses of biodiversity would be catastrophic for humanity and ecosystems as it would lead to the loss of ecosystem functions, the collapse of food webs, adversely affect human health, degrade medicine production and discovery, and more (Roe, Seddon, and Elliott 2018). The rate of biodiversity loss is unprecedented, and we are quickly moving towards a sixth mass extinction (Cowie, Bouchet, and Fontaine 2022). To combat biodiversity loss effectively, conservation and ecological rehabilitation efforts require accurate, robust models that can predict biodiversity patterns, identify priority areas, and assess potential conservation outcomes (Sinclair, White, and Newell 2010; Pollock et al. 2020).

To effectively predict biodiversity, it is crucial to have a clear understanding of how biodiversity data is collected and measured. Generally, biodiversity data is collected through field surveys, remote sensing, and citizen science initiatives and other monitoring schemes depending on what is being measured (Proença et al. 2017; Beery et al. 2021; Brun et al. 2024). These approaches or protocols can provide data on species occurrences, abundances, and distributions across different spatial and temporal scales (Proença et al. 2017). Common biodiversity datasets can be categorized by the type of species information they contain. Presence-only data record locations where a species has been observed, but do not provide information on confirmed absences. Presence-absence data record whether each species is present or absent at a site. Abundance data provide counts or estimates of the number of individuals of each species at a site. Once these raw data is collected, there are three broad categories of biodiversity metrics that are computed: alpha diversity, beta diversity, and gamma diversity. Alpha diversity refers to diversity within a site, such as the diversity within a specific community or ecosystem. Beta diversity refers to the diversity between different communities or ecosystems. Gamma diversity refers to the species diversity across communities or ecosystems within a larger region (Andermann et al. 2022). For the purposes of this thesis, we will focus on alpha diversity.

Common alpha diversity metrics include species richness, absolute abundance, relative abundance, Shannon index, arithmetic mean abundance, and geometric mean abundance (Santini et al. 2017; Beery et al. 2021). Species richness is the most commonly used metric and simply counts the number of species in a given area (Ferrier and Guisan 2006). Species richness is a very popular metric because presence-absence data is the most common type of biodiversity data (Özkan Tümer et al. 2026). Absolute abundance measures the total number of individuals of each species at a site, and relative abundance measures the proportion of individuals of each species relative to the total number of individuals in a site (Beery et al. 2021). The Shannon index is another commonly used metric and combines species richness and evenness into a single metric and quantifies the uncertainty in predicting the species of an individual randomly selected from a community (Beery et al. 2021). Arithmetic mean abundance is the average abundance of species in a given area, and geometric mean abundance is the exponential of the average of the log abundances of species in a given area (Santini et al. 2017).

For the purposes of this thesis, it is beneficial to express these terms mathematically. Biodiversity metrics are usually calculated at the site level. A site is a spatial sampling unit that is defined according to a sampling protocol. We define a site as r, which is a spatiotemporal location, and within a biodiversity dataset there are n sites which make up the region of interest $R = \{r_1, r_2, \ldots, r_n\}$. Across the entire region of interest R there is a set S of unique species that were observed, and s is a specific species, where $s \in S$. Furthermore, let a_{s,r_i} be the abundance or count of individuals of species s at site r_i.

The alpha diversity metrics used in this thesis are therefore defined at the site level as follows.

Species richness. Following Beery et al. (Beery et al. 2021), species richness is the number of unique species s at a given site r_i and can be expressed as:

$$SR_{r_i} = \sum_{s \in S} \mathbf{1}(a_{s,r_i} > 0), \tag{1}$$

where $\mathbf{1}(\cdot)$ is the indicator function that equals 1 if the condition is true and 0 otherwise. For reference, regional species richness across R can be expressed as:

$$SR_R = |S| \,. \tag{2}$$

Absolute abundance. Absolute abundance is the total number of individuals across species s at a given site r_i and can be expressed as:

$$A_{r_i} = \sum_{s \in S} a_{s,r_i}, \tag{3}$$

where a_{s,r_i} is the abundance or count of individuals of species s at site r_i (Beery et al. 2021), and $a_{s,r_i} = 0$ if species s is not present at site r_i.

Relative abundance. Relative abundance is the proportion of individuals of each species s relative to the total number of individuals at a given site r_i and can be expressed as:

$$p_{s,r_i} = \frac{a_{s,r_i}}{A_{r_i}}, \tag{4}$$

where a_{s,r_i} is the abundance or count of individuals of species s at site r_i and A_{r_i} is the total abundance of all species at site r_i (Beery et al. 2021). If $a_{s,r_i} = 0$ then species s is not present at site r_i, and $p_{s,r_i} = 0$.

Shannon index. The Shannon index combines species richness and evenness into a single metric and quantifies the uncertainty in predicting the species of an individual randomly selected from a community. It can be expressed as:

$$H_{r_i} = -\sum_{s \in S} p_{s,r_i} \ln(p_{s,r_i}), \tag{5}$$

where $p_{s,r_i}$is the relative abundance of species s at site r_i. If $p_{s,r_i} = 0$, then that term is not include in the calculation. The interpretation of the Shannon index is that higher values of H_{r_i} indicate higher diversity (Dušek and Popelková 2017).

Arithmetic mean abundance. Arithmetic mean abundance is the average abundance of species present at site r_i. It can be expressed as:

$$M_{r_i} = \frac{1}{SR_{r_i}} \sum_{s \in S} a_{s,r_i}, \tag{6}$$

where a_{s,r_i} is the abundance or count of individuals of species s at site r_i (Santini et al. 2017). If $a_{s,r_i} = 0$ for any species s in site r_i, then that species is not included in the calculation of the arithmetic mean abundance for site r_i.

Geometric mean abundance. Geometric mean abundance is the exponential of the average of the log abundances of species at a given site r_i and can be expressed as:

$$G_{r_i} = \exp\left(\frac{1}{SR_{r_i}} \sum_{s \in S} \ln\left(a_{s,r_i}\right)\right), \tag{7}$$

where a_{s,r_i} is the abundance or count of individuals of species s at site r_i (Santini et al. 2017). If $a_{s,r_i} = 0$ for any species s at site r_i, then that species is not included in the calculation of the geometric mean abundance for site r_i.

These metrics each have their own benefits and limitations. Species richness is the most commonly used metric because presence-absence data is the most common type of biodiversity data and it is easy to calculate (Lamb et al. 2009; Özkan Tümer et al. 2026). However, species richness does not account for the abundance of species and can be biased towards rare species (Ferrier and Guisan 2006). Absolute abundance and relative abundance capture more information about the abundance of species in a given area, but they can be dominated by common species (Beery et al. 2021). The Shannon index combines species richness and evenness into a single metric and can provide a more comprehensive measure of biodiversity, but it can be difficult to interpret (Dušek and Popelková 2017). Arithmetic mean abundance and geometric mean abundance are useful for measuring changes in overall populations but can be inflated by very high population density species (Santini et al. 2017). It is important to look at a wide range of metrics when measuring biodiversity as each metric measures different aspects of biodiversity and responds differently to changes in biodiversity (Lamb et al. 2009; Santini et al. 2017).

2.2 Spatial Biodiversity Modeling

Modeling biodiversity has historical roots in the early 20th century with work on modeling individual species distributions (Grinnell 1904). Since then models have expanded to include ecological models based on expert knowledge (Beery et al. 2021), parametric statistical models (Guisan, Edwards, and T. Hastie 2002), and machine learning models (Özkan Tümer et al. 2026). Spatial biodiversity models can be broadly divided into three strategies: bottom-up or "predict first, assemble later", joint or "assemble and predict together", and top-down or "assemble first, predict later' (Ferrier and Guisan 2006; Pollock et al. 2020). These strategies differ in how they produce predictions of community-level biodiversity metrics, and one of the key differences between top-down strategies and the other two is that top-down models directly predict community-level metrics. Top-down models directly output species richness or abundance, while the other two strategies first model the distribution of individual or multiple species and then aggregate those predictions to estimate species richness or abundance (Pollock et al. 2020). These top-down modeling approaches have been applied to predicting alpha, beta, and gamma biodiversity metrics across various spatial scales (Zhang et al. 2019; Hoskins et al. 2020; Mondal and Bhat 2021; Andermann et al. 2022).

Top-down modeling approaches come in a variety of forms, including regression-based models (Mondal and Bhat 2021), generalized dissimilarity models, which are specific to predicting beta diversity (Ferrier, Manion, et al. 2007; Zhang et al. 2019; Hoskins et al. 2020), and machine learning models (Andermann et al. 2022). To capture the broad range of top-down modeling approaches, we will define a mathematical framework that captures the shared structures of these models. Building on the notation from Section 2.1 and the formulations of Beery et al. (Beery et al. 2021) we define the following notation for a top-down modeling approach.

A site is denoted as $r \in \mathcal{R}$, where $\mathcal{R}$ is the spatiotemporal region of the study. Let $Y(r)$ denote the biodiversity response variable at site r. A general biodiversity dataset is defined as

$$\mathcal{D}_r = \{(r_i, y_i)\}_{i=1}^n,$$

where $r_i \in \mathcal{R}$ is a sampled site and y_i is the observed realization of $Y(r_i)$. The response variable depends on the community-level biodiversity metric being measured and described in Section 2.1. For example, species richness and abundance may be represented as non-negative counts, while diversity metrics may be represented as non-negative real-valued indices. We then extract a representation of the environment in the region $\mathcal{R}$ as

$$h : \mathcal{R} \to \mathcal{X} \subseteq \mathbb{R}^p.$$

Thus,

$$h(r_i) = x_i$$

where x_i is the vector of predictor variables for site r_i and $x_i \in \mathcal{X} \subseteq \mathbb{R}^p$, and p is the number of predictor variables. The predictor variables are usually gathered from various sources, such as remote sensing data, climate datasets, and topographic maps (Waltari et al. 2014; Beery et al. 2021). Therefore, the dataset can be represented as

$$\mathcal{D}_x = \{(x_i, y_i)\}_{i=1}^n.$$

We then define a model

$$f_\Theta : \mathcal{X} \to \mathbb{R},$$
$$\hat{y}_i = f_{\hat{\Theta}}(x_i) = f_{\hat{\Theta}}(h(r_i)),$$

with fitted parameters $\hat{\Theta}$, where $\hat{y}_i$ is the model-based approximation of the community-level biodiversity response $Y(r_i)$.. This general formulation allows for flexibility in the choice of model and the type of community-level biodiversity metric being predicted.

2.3 Models

This subsection reviews the different types of models that were used for top-down spatial biodiversity modeling in this master's thesis. The models discussed here are a subset of the models that can be used for top-down spatial biodiversity modeling. We define the assumptions and limitations of these models, as well as the different types of data they can be applied to. The models discussed include generalized linear models (Nelder and Wedderburn 1972), generalized additive models (T. J. Hastie and Tibshirani 1986), random forests (Breiman 2001b), gradient boosting (Friedman 2001), and neural networks (Bishop 2006). These models are used later in the methods to demonstrate the standardized pipeline for top-down spatial biodiversity modeling.

2.3.1 Generalized Linear Models

Generalized Linear Models (GLMs) are an extension of linear regression models that allow the response variable to follow a distribution from the exponential family (Nelder and Wedderburn 1972). Under normal linear regression, the response variable Y is modeled as

$$Y = \beta_0 + \beta_1 x_1 + \cdots + \beta_p x_p + \epsilon,$$

where ϵ is the error term. Linear regression assumes that the conditional mean of the response variable Y is linear in the predictors $x_1, x_2, \cdots, x_p$, the error term ϵ is normally distributed with mean 0 and variance σ^2 or $\epsilon \sim \mathcal{N}(0, \sigma^2)$, the error term ϵ is independent and identically distributed, and the variance of Y is constant across observations (Guisan, Edwards, and T. Hastie 2002).

In GLMs, the response variable for the i-th observation, Y_i, is assumed to follow a distribution from the exponential family, and a link function $g(\cdot)$ is used to connect the linear predictor to the mean of the response variable. The model is then specified as

$$g(\mathbb{E}(Y_i|\mathbf{x}_i)) = \beta_0 + \beta_1 x_{i1} + \cdots + \beta_p x_{ip},$$

where $\mathbb{E}(Y_i|\mathbf{x}_i) = \mu_i$ is the expected value of Y_i given the predictors $\mathbf{x}_i = (x_{i1}, x_{i2}, \ldots, x_{ip})$ and the linear predictor is $\eta_i = \beta_0 + \beta_1 x_{i1} + \cdots + \beta_p x_{ip}$ (Guisan, Edwards, and T. Hastie 2002). Then, we can rewrite the model as

$$g(\mu_i) = \eta_i.$$

To make predictions with a GLM, we use the inverse of the link function $g^{-1}(\cdot)$ to transform the linear predictor back to the scale of the response variable. The predicted value of the expected value for the i-th observation is then

$$\hat{\mu}_i = g^{-1}(\hat{\eta}_i) = g^{-1}(\hat{\beta}_0 + \hat{\beta}_1 x_{i1} + \cdots + \hat{\beta}_p x_{ip}).$$

Common link functions include the identity link for Gaussian response variables (Guisan, Edwards, and T. Hastie 2002) and a log link function for count data (Cai et al. 2023).

The main advantage of a GLM is that the response variable does not need to be normally distributed. Instead, GLMs allow the response to follow a distribution from the exponential family, such as Poisson models for count data and negative binomial models for overdispersed counts (Nelder and Wedderburn 1972; Cai et al. 2023), which provides greater flexibility for ecological data (Guisan, Edwards, and T. Hastie 2002). This is particularly useful for biodiversity metrics that are non-normal or overdispersed. For example, species richness is count data and often exhibits higher variance than expected under a Poisson model, making the negative binomial distribution a more appropriate choice (Cai et al. 2023). Another benefit of GLMs is that they are interpretable as the coefficients β_j represent the change in the linear predictor, or the transformed expected response $g(\mu_i)$, associated with a one-unit change in the predictor variable x_{ij} while holding all other predictors constant, and the interpretation depends on the choice of link function.

GLMs have been used as a top-down modeling strategy for predicting community-level biodiversity metrics (Brugere et al. 2023; Cai et al. 2023; Ngor et al. 2023). While GLMs are more flexible than linear regression models and have been shown to be effective for single species distribution modeling (Norberg et al. 2019), they can have limited performance when predicting community-level biodiversity metrics such as species richness (Robin and Hasif 2021; Ngor et al. 2023; Cai et al. 2023). First, GLM performance can be strongly influenced by the choice of predictor variables (Guisan, Edwards, and T. Hastie 2002). Second, GLMs may be insufficiently flexible when the true relationship between predictors and the response is highly nonlinear or complex (Yee and Mitchell 1991). Third, GLMs require the analyst to specify an appropriate response distribution, link function, and model structure, which should be chosen carefully to match the ecological context (Austin 2007). Although the latter limitations are often discussed in the context of species distribution modeling, they are also relevant to top-down models of community-level biodiversity metrics.

2.3.2 Generalized Additive Models

Generalized Additive Models (GAMs) are an extension of GLMs that allow for non-linear relationships between the predictors and the response variable (T. J. Hastie and Tibshirani 1986). In a GAM, the linear predictor is generalized to an additive predictor composed of smooth functions of the covariates. The model can be specified as

$$g(\mu_i) = \eta_i = \beta_0 + f_1(x_{i1}) + f_2(x_{i2}) + \cdots + f_p(x_{ip}),$$

where $\mu_i = \mathbb{E}(Y_i \mid \mathbf{x}_i)$, $g(\cdot)$ is the link function, and $f_j(\cdot)$ is a smooth function of the predictor variable x_{ij} (T. J. Hastie and Tibshirani 1986). The use of smooth functions in GAMs provides greater flexibility than GLMs, as they can capture complex, non-linear relationships between the predictors and the transformed expected response $g(\mu_i)$. The smooth functions themselves are very flexible and can be represented by many functions, such as splines, but are also commonly estimated from the data itself (T. J. Hastie and Tibshirani 1986; Yee and Mitchell 1991). The link function $g(\cdot)$ still must connect the expected response μ_i to the additive linear predictor η_i (T. J. Hastie and Tibshirani 1986).

The increased flexibility of GAMs has also contributed to their success in modeling individual species distributions (Valavi et al. 2022), and they have been utilized to predict community-level biodiversity metrics such as species richness and have shown better predictive performance than GLMs (Mondal and Bhat 2021; Cai et al. 2023). GAMs are also interpretable, as each smooth function f_j can be plotted as a function of its corresponding predictor variable x_{ij}, showing how the expected response changes across values of the predictor. However, some of the limitations of GAMs are similar to those of GLMs. First, GAM performance can also be strongly influenced by the choice of predictor variables (Guisan, Edwards, and T. Hastie 2002). While GAMs are more flexible than GLMs, this also increases model complexity and can increase computational cost (Mondal and Bhat 2021).

2.3.3 Random Forests

Random Forests are a supervised machine learning model that is based on the idea of combining several decision trees in an ensemble method to improve predictive performance (Breiman 2001b). To understand what a Random Forest is more concretely, we will first describe a decision tree. Given a set of predictor variables $X_1, X_2, \cdots, X_p$ and a response variable Y, a decision tree will recursively partition the predictor space into k regions such that the response variable is as homogeneous as possible within each region (Loh 2011). For a classification task with k classes, training a decision tree involves splitting the predictor space into regions $R_1, R_2, \cdots, R_m$ such that in each region R_j the predicted class is the majority class of the training observations that fall into that region (Moisen 2008; Loh 2011). For a regression task, the predicted value for each region R_j is the average of the response variable for the training observations that fall into that region (Moisen 2008; Loh 2011). The process of splitting the predictor space is done by searching over all predictor variables and candidate threshold values for each predictor. The best split is based on a criterion such as minimizing the variance or the mean squared error (MSE) (Klusowski 2020). For a regression tree, first, we would look at the sample variance or MSE of the node t, before the split, which is calculated as

$$MSE_t = \frac{1}{n_t}\sum_{i=1}^{n_t}(y_i - \bar{y}_t)^2,$$

where n_t is the number of observations, y_i is the response variable for the i-th observation in node t, and $\bar{y}_t$ is the mean among the observations in node t. Then, for a given split that divides the node t into two child nodes t_L and t_R with n_L and n_R observations, respectively, we would calculate the change in MSE. This can be expressed as

$$\Delta(MSE) = MSE_t - (\frac{n_L}{n_t}MSE_{t_L} + \frac{n_R}{n_t}MSE_{t_R}).$$

The MSE for the left and right children is calculated in the same manner as MSE_t but using the observations in the left and right children, respectively. The best split is the one that maximizes $\Delta(MSE)$ or equivalently minimizes the MSE of the two sub-samples proportionally (Klusowski 2020). This calculation is repeated for several potential splits across all predictor variables, and the best split is chosen based on this criterion. This process is repeated recursively on every subset until a stopping condition is met, such as a minimum number of observations in a leaf node or a maximum tree depth (Mienye and Jere 2024). An illustration of a decision tree can be seen in Figure 1.

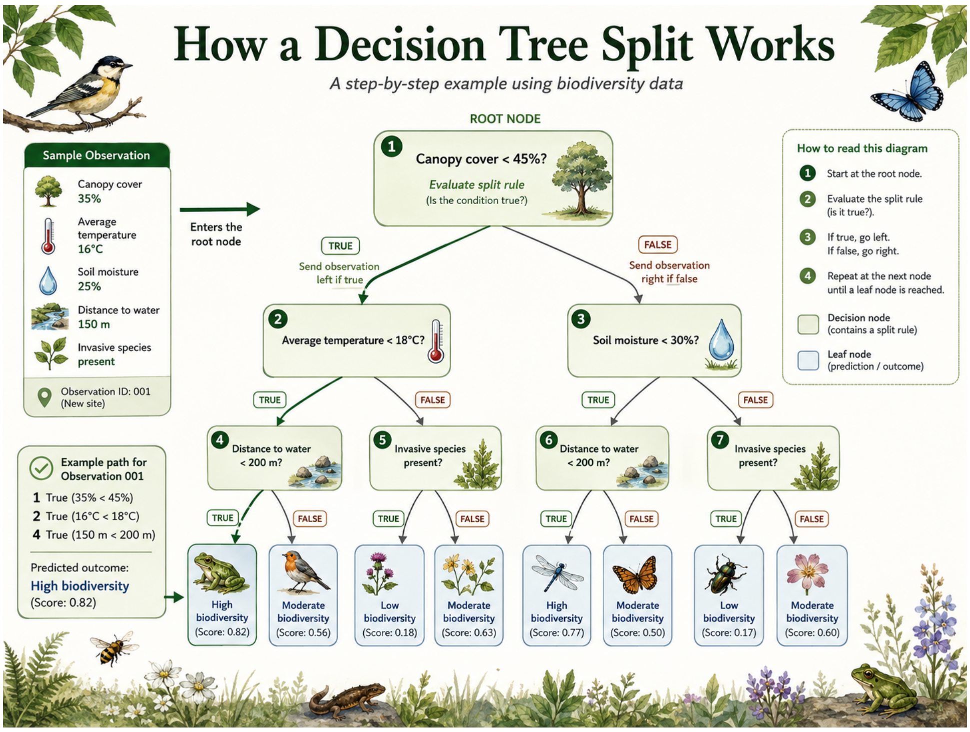

Figure 1: An illustrative decision tree for a regression task. The tree is built by recursively splitting the predictor space. Image generated with ChatGPT using the prompt: "Generate an image of a decision tree that shows how a split is processed. Use a biodiversity theme for the variables."

A Random Forest is an ensemble method that combines hundreds or thousands of decision trees to produce a prediction. Each tree is trained on a bootstrapped sample of the data, while each split considers only a random subset of the predictor variables. For regression tasks, the final prediction is obtained by averaging the predictions across all trees; for classification tasks, it is typically determined by majority vote (Breiman 2001b). This bootstrapping and aggregation, commonly referred to as bagging, also helps reduce the variance of the final model (Aria, Cuccurullo, and Gnasso 2021).

Random Forests are a popular machine learning model for top-down spatial biodiversity modeling (Olaya-Marín, Martínez-Capel, and Vezza 2013; Zhang et al. 2019; Večeřa et al. 2019; Ngor et al. 2023; Cai et al. 2023; Brugere et al. 2023; Baggström et al. 2025) due to their ability to capture complex, non-linear relationships between predictors and the response variable, and their robustness to overfitting due to their ensemble nature (Breiman 2001b). However, some of their shortcomings include being less interpretable than GLMs and GAMs because they are an ensemble of many, possibly deep, trees (Aria, Cuccurullo, and Gnasso 2021). Also, random forests can still show a gap between training and test performance (Olaya-Marín, Martínez-Capel, and Vezza 2013; Baggström et al. 2025). Furthermore, they can become computationally expensive when the number of trees is large or when the depth of the trees is large.

2.3.4 Gradient Boosting

2.3.5 Neural Networks

3 Methods/setup

3.1 Pipeline for Community Level Modeling

3.2 Data

3.3 Model Selection

4 Results/ Case Study

5 Discussion

5.1 Findings

5.2 Ethical Discussion

6 Future Work/ Applications

Future Work includes a full comprehensive benchmark of community level models across a wide range of datasets and models. The integration with GBIF should should allow for easy facilitation of this. This would allow for a more systematic evaluation of community level models and provide insights into their strengths and weaknesses in predicting biodiversity patterns and trends. Furthermore, a comprehensive benchmark would allow for a more nuanced evaluation of community-level models and their applicability across different ecological contexts, including comparisons across taxonomic groups, spatial scales, and environmental predictor sets. This would ultimately contribute to the advancement of biodiversity modeling as a whole and inform conservation strategies and policy decisions.

Talk about comparisons of top-down methods with bottom-up or assemble and predict together as future works. There are some in the Zotero on SDM vs. BDMs folder.

7 Conclusion

A Appendix A: Github Repository

Includes the instructions there

B Appendix B: Additional Results

Includes the additional results there

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